

Swaroop S. Hegde

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Research Interests

Additive Combinatorics, Ramsey Theory, Probabilistic Methods in Combinatorics

Education

University of Georgia 2021 - present

PhD in Mathematics Athens, GA

Advisor: Giorgis Petridis

National Institute of Science Education and Research (NISER) 2015 - 2020

Integrated M.Sc. in Mathematics Bhubaneswar, India

Advisor: Brundaban Sahu

- Master's Thesis: Combinatorics and Representation Theory of Symmetric Groups

Work Experience

Graduate Instructor, University of Georgia Aug 1, 2021 - present

Responsible for most aspects of the following courses including preparing course notes, delivering lectures, writing exams and quizzes, holding office hours and managing grades.

- Calculus I (Math 2250): Spring 2025, Fall 2024, Fall 2023, Spring 2023, Fall 2022
- PreCalculus (Math 1113): Fall 2026 (x2), Spring 2026 (x2), Fall 2025 (x2), Spring 2024 (x2), Spring 2022 (x2), Fall 2021
- Online Precalculus (Math 1113E): Summer 2024

Publications

Refined upper bounds on Schur-like numbers Aug 2026

S. Hegde, A. Lott, G. Petridis, N. R. Ponagandla

arXiv preprint: [arXiv:2608.03661](https://arxiv.org/abs/2608.03661)

Extensions of the Furstenberg-Sárközy theorem via the arithmetic level-d inequality May 2026

C. F. E. Adajar, R. Agrawal, M. R. Choudhuri, C. Y. Chuah, S. Fan, S. Hegde, A. Lott, K. Nandakumar, and N. R. Ponagandla

arXiv preprint: [arXiv:2605.16216](https://arxiv.org/abs/2605.16216)

An Inverse Result for Ruzsa's Inequality on Triple Sumsets Oct 2025

S. Hegde

arXiv preprint: [arXiv:2510.20073](https://arxiv.org/abs/2510.20073)

Awards

David Galewski Outstanding Graduate Teaching Award	Apr 1, 2024
Mathematics Department, University of Georgia	
Outstanding Teaching Assistant Award	Mar 1, 2024
Center for Teaching and Learning, University of Georgia	
Certificate of Merit	Sep 1, 2018
School of Mathematical Sciences, NISER — For outstanding performance in the Spring 2018 semester.	
INSPIRE Scholarship Award	Jan 1, 2015
Department of Science and Technology, Govt. of India	

Selected Talks

Generalized Schur Numbers	Sep 2026
UGA Analysis Seminar	
An Inverse Result for Ruzsa's Inequality on Triple Sumsets	Mar 2026
Graduate Student Combinatorics Conference, University of Illinois at Chicago	
An Inverse Result for Ruzsa's Inequality on Triple Sumsets	Oct 2025
Discrete Math Seminar, Iowa State University (virtual)	
An Inverse Result for Ruzsa's Inequality on Triple Sumsets	May 2025
CanADAM Conference, Ottawa, Canada	
An Inverse Result for Ruzsa's Inequality on Triple Sumsets	May 2025
Integers Conference, UGA	
Ramsey number $R(4,k)$	Mar 2024
Oral Qualifying Exam Presentation, UGA	
AKS Algorithm for Primality Testing	Oct 2023
Combinatorial Number Theory Reading Seminar, UGA	
A Polygon Problem	Oct 2023
Graduate Student Seminar, UGA	
The RSK Correspondence	Apr 2023
Graduate Student Seminar, UGA	
Almost Periods of Convolutions and Application to Roth's Theorem	Feb 2023
Additive Combinatorics Reading Seminar, UGA	

Cores and Quotients of Integer Partitions	Nov 2022
Graduate Student Seminar, UGA	
Rédei's Theorem and Szőnyi Method	Sep 2021
Polynomial Methods Reading Seminar, UGA	
Roth's Theorem via Finite Fourier Analysis	Apr 2019
Math Seminar, NISER	
Pebbling the Chessboard	Apr 2018
NISER Open Day	
The Story of Dimension	Feb 2017
MathematiX Club, NISER	
Clock Arithmetic	Apr 2016
NISER Open Day (for high school students)	
Gödel's Incompleteness Theorems	Nov 2015
MathematiX Club, NISER	
Conferences and Workshops	
Graduate Workshop in Logic and Combinatorics	Aug 2026
University of Illinois at Chicago	
International Conference on Graph Theory and its Applications (ICGTA 2026)	Jun 2026
Presidency University, Bangalore, India	
Simons School and Workshop on Harmonic Analysis and applications to Ramsey theory	Jun 2026
Rényi Institute, Budapest, Hungary	
Simons School and Workshop on Discrete Harmonic Analysis and Analytic Number Theory	May 2026
Rényi Institute, Budapest, Hungary	
Graduate Student Combinatorics Conference	Mar 2026
University of Illinois at Chicago	
PCMI 2025 Graduate Summer School on Extremal Combinatorics and Probabilistic Methods	Jul 2025
PCMI, Park City, Utah	
Graduate Research Workshop in Combinatorics	Jun 2025
Iowa State University, Ames, Iowa	
CanadAM Conference 2025	May 2025

Ottawa, Ontario, Canada

Integers Conference 2025 May 2025

University of Georgia, Athens, GA

Algebraic and Analytic Methods in Combinatorics Mar 2025

SLMath, Berkeley, CA

Introductory Workshop - Graph Theory: Extremal, Probabilistic and Structural Feb 2025

SLMath, Berkeley, CA

Early-Career Conference in Combinatorics Jun 2024

University of Illinois Urbana-Champaign, IL

33rd Cumberland Conference on Combinatorics, Graph Theory & Computing May 2024

Mississippi State University, MS

Formalization of Mathematics Jun 2023

SLMath, Berkeley, CA

Integers Conference 2023 May 2023

University of Georgia, Athens, GA

ATM-Schools AIS on Representation Theory of Finite Groups Jun 2017

Chennai Mathematical Institute, Chennai, India

Madhava Mathematics Camp Jun 2017

Indian Statistical Institute, Bangalore, India

National Science Camp (VIJYOSHI) Dec 2015

Kolkata, India

Selected Coursework

Selected Graduate Courses: Entropy Methods in Combinatorics, Ramsey Theory, Additive Combinatorics I & II, Probabilistic Methods in Combinatorics, Ergodic Theory, The Circle Method, Equations with Prime Variables, Algebraic Graph Theory, Quiver Representations, Commutative Algebra, Lie Algebras, Hopf Algebras

Selected Reading Courses: Asymptotics of Ramsey Number $R(4,t)$ (Oral Exam), Kelley-Meka bound for Roth's theorem, Some recent improvements on Freiman's Theorem, Polynomial Methods in Combinatorics, Reflection Groups

Mentoring

Directed Reading Program, UGA Math Department Jun 2021 - present

Mentored UGA undergraduate students on various projects

- Yan Heng: Formalization of Mathematics in LEAN, Spring 2026

- Benjamin Wang-Tie: Applications of Entropy in Combinatorics, Spring 2025
- Aakash Khanal: Mathematics of Card Shuffling, Summer 2021

Service

Volunteer for Sonia Kowalesky Day	Apr 25, 2026
University of Georgia	
Co-organizer of the Graduate Student Seminar	Jan 1, 2025 - May 1, 2025
Mathematics Department, University of Georgia	
Co-facilitator of the Precalculus community working group	Jan 1, 2024 - May 1, 2024
Mathematics Department, University of Georgia	
Volunteer for High School Math Tournament	Oct 1, 2023 - Oct 1, 2024
University of Georgia – Volunteered in Oct 2023 and Oct 2024.	
Peer Mentor for first year graduate students	Aug 1, 2023 - May 1, 2025
Mathematics Department, University of Georgia	
Secretary of Student Chapter of AMS	Aug 1, 2022 - May 1, 2023
Mathematics Department, University of Georgia	